Wireless Charging Facility Deployment and Adaptive Battery Configuration under Service-Plan Uncertainty

Author Name[a]

[a]*Affiliation*

Abstract

This abstract will be written after the main body is stable.

Keywords: Wireless charging, electric bus, stochastic programming, integer L-shaped method, Benders decomposition

1. Introduction

Electrifying bus fleets requires coordinated planning of charging infrastructure, battery capacity, and charging-resource usage under future service requirements. Wireless charging technologies expand the set of charging opportunities by allowing buses to receive energy at selected stops or road links during regular service. This capability turns infrastructure planning from a stand-alone siting decision into a joint planning problem involving wireless facility deployment, route-level battery configuration, and the use of existing plug-in fast chargers.

Wireless charging facilities are long-lived assets, whereas transit service plans may change with future demand, fleet composition, and operating policies. A layout that is attractive for one future service plan may perform poorly under another because buses may operate different trip chains, require different battery capacities, and compete for existing fast-charging resources at different times. The infrastructure decision therefore has to be made before the future fleet-service configuration is known, whereas battery-capacity and charging-resource decisions can adapt after that configuration is realized.

The central planning trade-off is shaped by the interaction among wireless infrastructure, battery capacity, and plug-in fast-charging capacity. More in-route wireless charging can reduce battery requirements on some routes and shift energy replenishment away from terminals or depots. At the same time, wireless deployment changes the timing and amount of residual plug-in fast charging needed by different routes. Because plug-in fast chargers are limited shared resources, these residual demands couple route-level decisions that would otherwise be separable.

We capture this timing and coupling structure with a two-stage stochastic mixed-integer planning model. The first stage selects stationary and dynamic wireless charging facilities. The second stage, after a fleet-service scenario is realized, chooses scenario-adaptive battery configurations and representative-day charging schedules. The resulting recourse problems contain discrete battery choices, charging-scheduling decisions, and shared fast-charger capacity constraints. Consequently, both the deterministic equivalent and the classical integer L-shaped decomposition become difficult to scale as the network and scenario set grow.

To solve this model, we develop a reformulation-enabled integer L-shaped method. The method differs from a standard integer L-shaped decomposition by changing what recourse information is exposed to the master and by separating inexpensive lower-bound generation from exact recourse verification. This design preserves exactness when the cut loop converges and creates the algorithmic components evaluated in the computational study.

The paper makes three contributions. First, it identifies a two-stage stochastic mixed-integer structure in which here-and-now facility decisions interact with scenario-dependent discrete design choices and shared-resource operational recourse. The wireless-charging application provides a concrete instantiation of this structure, and the formulation highlights the coupling that drives the algorithmic development. Second, it develops an exact reformulation-enabled integer L-shaped algorithm that combines scenario-dependent lifting, a two-phase relaxation-verification scheme, and monotonicity-strengthened cuts for mixed-integer recourse. Third, it provides a computational performance analysis that evaluates scalability with respect to network size and scenario count, and tests the method's adaptability when the gap between the Phase 1 relaxation and Phase 2 exact verification varies.

The experiments use wireless-charging planning instances constructed from Singapore bus-service data for routes operated by Tower Transit. They are designed to answer two questions: whether the model produces interpretable planning decisions, and which algorithmic components expand the solvable instance boundary. They also vary the available plug-in fast-charging capacity to test how the proposed method behaves as shared-resource coupling becomes tighter or looser.

The remainder of the paper is organized as follows. Section 2 reviews related work on electric-bus charging infrastructure planning and stochastic decomposition methods. Section 3 presents the two-stage formulation and the baseline integer L-shaped decomposition. Section 4 develops the lifted two-phase algorithm and strengthened cuts. Section 5 reports the computational results. Section 6 concludes.

2. Literature Review

This section reviews two streams of literature that define the paper's application structure and algorithmic positioning. The first stream is electric-bus charging and wireless infrastructure planning, which motivates the interaction among wireless facilities, battery choices, and existing plug-in fast-charging resources. The second stream is stochastic facility-location modeling and decomposition, which provides the methodological context for two-stage mixed-integer planning with scenario-dependent recourse. The review is organized to keep the transportation discussion concise and to focus on the shared-resource mixed-integer recourse structure used by the proposed algorithm.

2.1. Electric-bus charging and wireless infrastructure planning

Electric-bus charging studies commonly coordinate infrastructure deployment, battery sizing, vehicle or charging schedules, and service feasibility. These decisions are often formulated as integrated optimization models because charging resources, route energy requirements, and schedule feasibility interact. Depot and plug-in fast-charging studies emphasize charger availability and charging schedules, whereas opportunity-charging studies add charging possibilities during service.

Wireless charging infrastructure planning extends this setting by considering stationary wireless charging at stops and dynamic wireless charging on road links. This literature provides modeling elements that are central to our application: candidate wireless facilities, route-level energy feasibility, battery-capacity implications, and charging-operation constraints. It also motivates the coexistence of wireless facilities with existing plug-in fast-charging resources.

Our use of this stream is structural. We adopt wireless-facility placement, route-level battery choices, representative-day charging operations, and existing plug-in fast chargers as the application objects of the model. We do not focus on real-time charging control, bus dispatch, or traffic operations; instead, these objects define a long-term planning setting whose shared-resource mixed-integer recourse structure is reviewed methodologically next.

2.2. Stochastic facility-location models and decomposition methods

Stochastic facility-location and network-design models provide the optimization background for here-and-now infrastructure decisions under uncertain future conditions. In these models, scenarios typically represent uncertain demand, service requirements, or operating conditions, and the recourse problem evaluates the consequence of a first-stage design after a scenario is realized. Benders decomposition, integer L-shaped methods, and logic-based Benders decomposition provide the standard decomposition language for such two-stage models, especially when the deterministic equivalent is too large or the recourse problem contains discrete decisions.

Within this decomposition family, computational performance is shaped by two broad design dimensions. The first is the decomposition or reformulation choice: what information is retained in the master problem, what remains in the scenario subproblems, and whether the subproblems can be decomposed or relaxed without losing useful bound information. The second is the separation strategy: how feasibility and optimality information from the subproblems is converted into cuts, and whether those cuts exploit problem-specific structure beyond a basic no-good exclusion.

Research on decomposition for mixed-integer recourse has developed techniques along both dimensions. On the reformulation side, stronger master representations, partial recourse descriptions, relaxed subproblems, and internal subproblem decompositions can improve lower bounds or reduce separation cost. On the cut side, integer L-shaped cuts, logic-based Benders cuts, combinatorial cuts, valid inequalities, and dominance-based cuts can exploit structure that is not captured by generic linear duality. The effectiveness and validity of these enhancements are structure dependent, which is why successful decomposition algorithms are often tailored to the recourse model rather than transferred directly from one application to another.

This paper follows that tailored-decomposition perspective. The electric-bus wireless-charging setting creates a shared-resource mixed-integer recourse structure with scenario-dependent battery choices, route-level operating decisions, and plug-in fast-charger capacity coupling. The proposed method instantiates the two design dimensions above through a reformulation-enabled integer L-shaped algorithm: it changes the master–subproblem organization, uses a relaxation–verification separation scheme, and derives cuts from the joint facility–battery structure while retaining exact recourse verification.

3. Problem Formulation and Recourse Structure

This section formulates the wireless charging infrastructure planning problem as a two-stage stochastic program. The formulation separates long-lived wireless-facility decisions from scenario-adaptive battery and charging decisions, and then identifies the recourse structure used by the algorithm in Section 4.

3.1. Two-stage stochastic formulation

Let Ω be the set of service-plan scenarios, and let p_ω be the probability of scenario $\omega \in \Omega$, where $p_\omega > 0$ and $\sum_{\omega\in\Omega} p_\omega = 1$. Each scenario represents one plausible future service plan and fixes the route frequencies, timetables, bus trip chains, active-fleet requirements, and trip-level energy requirements, while the operated route set $\mathcal{R}$ is common across scenarios. The planner first installs long-lived wireless charging facilities before the future service plan is known. After a scenario is realized, the second-stage recourse selects route-level battery capacities and the corresponding representative-day charging allocation and energy-management decisions.

Let $\mathcal{S}$ be the station set, $\mathcal{L}$ the directed link set between consecutive stations, and $C \subseteq \mathcal{S}$ the set of existing interchanges or hubs equipped with plug-in chargers. Stationary wireless charging (SWC) can be installed at any station in $\mathcal{S} \setminus C$, while dynamic wireless charging (DWC) can be installed on any directed link in $\mathcal{L}$. The first-stage binary variables w_s^{S} and w_l^{D} indicate whether SWC is installed at station $s \in \mathcal{S} \setminus C$ and DWC is installed on link $l \in \mathcal{L}$, respectively. Existing plug-in chargers at stations in C are pre-existing resources with limited shared capacity and are not candidates for SWC deployment.

The compact two-stage stochastic formulation is

$$\min_{w^{\mathrm{S}},w^{\mathrm{D}}} \quad C^{\mathrm{fac}}(w) + \sum_{\omega\in\Omega} p_\omega \Phi_\omega(w) \tag{1}$$

$$\text{s.t.} \quad w_s^{\mathrm{S}} \in \{0,1\}, \qquad s \in \mathcal{S} \setminus C, \tag{2}$$

$$w_l^{\mathrm{D}} \in \{0,1\}, \qquad l \in \mathcal{L}. \tag{3}$$

where $w = (w^{\mathrm{S}}, w^{\mathrm{D}})$ and

$$C^{\mathrm{fac}}(w) = \sum_{s\in\mathcal{S}\setminus C} c_s^{\mathrm{S}} w_s^{\mathrm{S}} + \sum_{l\in\mathcal{L}} c_l^{\mathrm{D}} w_l^{\mathrm{D}}.$$

Here c_s^{S} and c_l^{D} are the representative-day equivalent costs of installing SWC at station s and DWC on link l, respectively. Facility and battery investment costs are converted to representative-day equivalents to match the charging-cost scale. The vector w is the nonanticipative here-and-now facility decision, and $\Phi_\omega(w)$ is the optimal second-stage value in scenario ω.

The scenario decisions consist of route-level battery selection variables $z_{\omega,r,k}$, plug-in charger occupancy variables collected in x_ω, and remaining operating variables collected in y_ω, including charging amounts and battery energy states. For a fixed facility layout w, the scenario recourse value $\Phi_\omega(w)$ is the minimum scenario cost

$$\Phi_\omega(w) = \min_{z_\omega,x_\omega,y_\omega} \left\{ C_\omega^{\mathrm{bat}}(z_\omega) + C_\omega^{\mathrm{chg}}(y_\omega) : (z_\omega, x_\omega, y_\omega) \in \mathcal{X}_\omega(w) \right\}. \tag{4}$$

The next subsection expands $\mathcal{X}_\omega(w)$.

3.2. Scenario recourse formulation

For a fixed scenario ω, the recourse formulation uses route-indexed trip-chain data. For each route $r \in \mathcal{R}$, $\mathcal{M}_{\omega,r}$ is the set of trip chains, $\mathcal{A}_{\omega,r}$ is the set of directed traversals, and $\mathcal{V}_{\omega,r}^{\mathrm{P}}$ and $\mathcal{V}_{\omega,r}^{\mathrm{S}}$ are the plug-in and SWC station visits. Each trip chain corresponds to one bus, so $n_{\omega,r} = |\mathcal{M}_{\omega,r}|$ is the route fleet size in scenario ω. A consolidated notation table is provided in Appendix Appendix A.

Recourse costs.. Battery options $\mathcal{K}$ have capacities β_k, with representative-day equivalent unit cost c^{bat}. Operating cost is based on grid-side energy purchased through plug-in charging, stationary wireless charging, dynamic wireless charging, and overnight depot charging. The corresponding charging amounts are q_v^{P}, q_v^{S}, q_a^{D}, and q_m^{N}, with nonnegative daytime and overnight electricity prices c^{day} and c^{night}. The scenario battery cost is

$$C_\omega^{\mathrm{bat}}(z_\omega) = c^{\mathrm{bat}} \sum_{r\in\mathcal{R}} n_{\omega,r} \sum_{k\in\mathcal{K}} \beta_k z_{\omega,r,k}.$$

The scenario charging cost is

$$C_\omega^{\mathrm{chg}}(y_\omega) = c^{\mathrm{day}} \left(\sum_{r\in\mathcal{R}} \sum_{v\in\mathcal{V}_{\omega,r}^{\mathrm{P}}} q_v^{\mathrm{P}} + \sum_{r\in\mathcal{R}} \sum_{v\in\mathcal{V}_{\omega,r}^{\mathrm{S}}} q_v^{\mathrm{S}} + \sum_{r\in\mathcal{R}} \sum_{a\in\mathcal{A}_{\omega,r}} q_a^{\mathrm{D}} \right) + c^{\mathrm{night}} \sum_{r\in\mathcal{R}} \sum_{m\in\mathcal{M}_{\omega,r}} q_m^{\mathrm{N}}.$$

Battery configuration.. The route-level battery-choice constraints are

$$\sum_{k\in\mathcal{K}} z_{\omega,r,k} = 1, \qquad r \in \mathcal{R}, \tag{5}$$

$$B_{\omega,r} = \sum_{k\in\mathcal{K}} \beta_k z_{\omega,r,k}, \qquad r \in \mathcal{R}, \tag{6}$$

$$z_{\omega,r,k} \in \{0,1\}, \qquad r \in \mathcal{R},\ k \in \mathcal{K}. \tag{7}$$

Constraints (5)–(7) select one common battery option for all buses assigned to each route and define the induced route battery capacity $B_{\omega,r}$.

Shared plug-in charging.. Plug-in charger use is modeled over scenario time slots $\mathcal{T}_\omega$. A visit v can use plug-in charging only during $\mathcal{T}_v \subseteq \mathcal{T}_\omega$, and $s(v)$ maps the visit to its station. The binary variable $x_{v\tau}$ records whether visit v occupies a plug-in charger in time slot τ; each occupied slot delivers up to κ units of grid-side energy, and station s has capacity U_s. The plug-in charging constraints are

$$0 \le q^{\mathrm{P}}_v \le \kappa \sum_{\tau\in\mathcal{T}_v} x_{v\tau}, \qquad r \in \mathcal{R},\ v \in \mathcal{V}^{\mathrm{P}}_{\omega,r}, \tag{8}$$

$$\sum_{r\in\mathcal{R}} \sum_{\substack{v\in\mathcal{V}^{\mathrm{P}}_{\omega,r}:\\ s(v)=s,\ \tau\in\mathcal{T}_v}} x_{v\tau} \le U_s, \qquad s \in C,\ \tau \in \mathcal{T}_\omega, \tag{9}$$

$$x_{v\tau} \in \{0,1\}, \qquad r \in \mathcal{R},\ v \in \mathcal{V}^{\mathrm{P}}_{\omega,r},\ \tau \in \mathcal{T}_v. \tag{10}$$

Constraints (8)–(10) define plug-in charging amounts, plug-in charger occupancy, and station-time capacity. The shared plug-in charger-capacity constraint (9) provides the main cross-route coupling within a scenario.

Route-local operational feasibility.. Route energy feasibility is written over station visits $\mathcal{V}_{\omega,r}$ and directed traversals $\mathcal{A}_{\omega,r}$. The mappings $\ell(a)$, tail(a), and head(a) connect each traversal to its physical link and adjacent visits. Traversal and depot energy requirements are represented by d_a, d^0_m, and d^N_m, while $\bar{q}$ and η collect charging-opportunity limits and charging efficiencies. The variables e^{arr}_v and e^{dep}_v are the battery energy states at arrival and departure of visit v. The state-of-charge fractions are bounded by $\underline{\rho}$ and $\overline{\rho}$, and v^0_m and v^N_m denote the first and last visits of trip chain m; detailed notation is listed in Appendix Appendix A. Trip-level energy requirements are exogenous and do not vary with the selected battery capacity. Overnight depot charging is assumed to have sufficient power and capacity to restore all buses to the prescribed initial state of charge before the next representative day. Event-level variables suppress scenario and route indices because each visit and traversal is already a scenario-route-specific copy. Charging quantities are nonnegative continuous grid-side energy variables, and battery-energy states are continuous. For each route $r \in \mathcal{R}$, the route-local operational constraints are

$$0 \le q^{\mathrm{S}}_v \le \bar{q}^{\mathrm{S}}_v w^{\mathrm{S}}_{s(v)}, \qquad v \in \mathcal{V}^{\mathrm{S}}_{\omega,r}, \tag{11}$$

$$0 \le q^{\mathrm{D}}_a \le \bar{q}^{\mathrm{D}}_a w^{\mathrm{D}}_{\ell(a)}, \qquad a \in \mathcal{A}_{\omega,r}. \tag{12}$$

$$\underline{\rho} B_{\omega,r} \le e^{\mathrm{arr}}_v \le \overline{\rho} B_{\omega,r}, \qquad v \in \mathcal{V}_{\omega,r}, \tag{13}$$

$$\underline{\rho} B_{\omega,r} \le e^{\mathrm{dep}}_v \le \overline{\rho} B_{\omega,r}, \qquad v \in \mathcal{V}_{\omega,r}, \tag{14}$$

$$e^{\mathrm{dep}}_v = e^{\mathrm{arr}}_v + \eta^{\mathrm{P}} q^{\mathrm{P}}_v, \qquad v \in \mathcal{V}^{\mathrm{P}}_{\omega,r}, \tag{15}$$

$$e^{\mathrm{dep}}_v = e^{\mathrm{arr}}_v + \eta^{\mathrm{S}} q^{\mathrm{S}}_v, \qquad v \in \mathcal{V}^{\mathrm{S}}_{\omega,r},$$

$$e^{\mathrm{arr}}_{\mathrm{head}(a)} = e^{\mathrm{dep}}_{\mathrm{tail}(a)} - d_a + \eta^{\mathrm{D}} q^{\mathrm{D}}_a, \qquad a \in \mathcal{A}_{\omega,r}, \tag{16}$$

$$e^{\mathrm{arr}}_{v^0_m} = \overline{\rho} B_{\omega,r} - d^0_m, \qquad m \in \mathcal{M}_{\omega,r}, \tag{17}$$

$$e^{\mathrm{dep}}_{v^N_m} - d^N_m \ge \underline{\rho} B_{\omega,r}, \qquad m \in \mathcal{M}_{\omega,r}, \tag{18}$$

$$\eta^{\mathrm{P}} q^{\mathrm{N}}_m = \overline{\rho} B_{\omega,r} - e^{\mathrm{dep}}_{v^N_m} + d^N_m, \qquad m \in \mathcal{M}_{\omega,r}. \tag{19}$$

Constraints (11)–(12) link route operations to first-stage wireless-facility decisions, and constraints (13)–(14) impose arrival and departure state-of-charge bounds. Constraints (15)–(16) update the energy state through station charging and directed traversals, while constraints (17)–(19) close each trip chain through depot movements and overnight recharge.

3.3. Recourse structure

For a fixed facility layout w, the expected recourse term is separable across scenarios. This yields a standard integer L-shaped reference in which each scenario is evaluated as a monolithic mixed-integer recourse problem over $(z_\omega, x_\omega, y_\omega)$. We refer to this direct decomposition as Natural ILS. The recourse problem also exhibits conditional route locality. Constraints (11)–(19) are imposed independently by route, whereas the shared plug-in charger-capacity constraint (9) couples plug-in occupancy across routes at each station–time pair. Finally, the formulation induces a facility–battery resource order. Under the model specification that trip-level energy consumption is independent of the selected battery capacity, additional wireless facilities and larger route batteries preserve operational feasibility under resource-dominating configurations and cannot increase the minimum charging cost.

These properties define a layered recourse structure comprising scenario-dependent discrete battery configurations, conditionally route-local operations, and limited shared-capacity coupling. Section 4 develops an exact decomposition method tailored to this structure.

4. Reformulation-Enabled Integer L-Shaped Method

This section develops an exact integer L-shaped method that exploits the recourse structure identified in Section 3. The method lifts scenario-dependent battery configurations into the decomposition, derives lower bounds from route-level LP relaxations, verifies candidate configurations through exact mixed-integer operational subproblems, and strengthens scenario-level pointwise cuts using the model-induced facility–battery resource order.

4.1. Lifted value-function reformulation

In a natural integer L-shaped decomposition of (1), only the facility variables w are explicit among the discrete configuration decisions, while each scenario recourse evaluation jointly optimizes the battery configuration z_ω and the operational decisions (x_ω, y_ω). This aggregation keeps the battery choice embedded in the mixed-integer recourse problem. Consequently, the master-side support of the integer L-shaped cuts is restricted to the facility variables. We therefore make z_ω explicit in the lifted decomposition. Conditional on (w, z_ω), the remaining recourse contains only the operational decisions, and cuts can be generated in the joint facility–battery configuration space. The facility layout w remains a here-and-now decision, whereas z_ω and (x_ω, y_ω) remain scenario-adaptive decisions made after scenario ω is realized.

To make this separation explicit, we first condition on a candidate facility–battery configuration. For a fixed (w, z_ω), the route battery capacities are fixed through (6), and the remaining scenario evaluation is the exact operational MIP

$$\begin{aligned} Q_\omega(w, z_\omega) = \min_{x_\omega, y_\omega} \quad & C_\omega^{\mathrm{chg}}(y_\omega) \\ \text{s.t.} \quad & (8)\text{–}(10), \\ & (11)\text{–}(19), \quad r \in \mathcal{R}. \end{aligned} \tag{20}$$

The first constraint group retains scenario-level plug-in occupancy and shared-capacity constraints, while the second contains all route-indexed operational constraints. If this MIP is infeasible, we set $Q_\omega(w, z_\omega) = +\infty$.

With Q_ω defined in this way, the scenario value in (4) satisfies

$$\begin{aligned} \Phi_\omega(w) = \min_{z_\omega} \quad & C_\omega^{\mathrm{bat}}(z_\omega) + Q_\omega(w, z_\omega) \\ \text{s.t.} \quad & (5)\text{–}(7). \end{aligned} \tag{21}$$

Let $z = (z_\omega)_{\omega \in \Omega}$ collect the scenario-dependent battery-selection vectors. Substituting (21) into the original two-stage formulation (1) yields the following exact lifted value-function formulation:

$$\begin{aligned} V^L = \min_{w, z} \quad & \left[C^{\mathrm{fac}}(w) + \sum_{\omega \in \Omega} p_\omega C_\omega^{\mathrm{bat}}(z_\omega) \right] + \sum_{\omega \in \Omega} p_\omega Q_\omega(w, z_\omega) \\ \text{s.t.} \quad & (2)\text{–}(3), \\ & (5)\text{–}(7), \qquad \omega \in \Omega. \end{aligned} \tag{22}$$

In this formulation, both the facility variables w and the scenario-dependent battery variables z_ω are explicit in the decomposition, whereas the remaining operational decisions are represented through the conditional value function $Q_\omega(w, z_\omega)$. Thus, (22) is an exact reformulation of (1): it changes the decomposition structure but not the decision timing, information structure, feasible solution set, or optimal value.

4.2. Two-phase lower bounding and exact separation

The lifted reformulation makes the scenario battery variables explicit, but exact lifted integer L-shaped separation remains computationally demanding. For each candidate (w, z_ω), exact scenario evaluation must solve an operational MIP with route coupling from shared plug-in capacity and binary plug-in occupancy. The two-phase method first solves a lower-bounding formulation with route-separable LP recourse and then uses the information generated in Phase 1 to initialize exact lifted integer L-shaped separation.

The operating vector decomposes by route as $y_\omega = (y_{\omega,r})_{r\in\mathcal{R}}$, and the charging cost is route additive. Phase 1 removes the shared plug-in charger-capacity constraint family (9). Once this coupling family is removed, the binary occupancy variables can be projected out exactly because they appear only in (8)–(10) and not in the objective. For each route r, the projection replaces these constraints with

$$0 \le q^{\mathrm{P}}_v \le \kappa|\mathcal{T}_v|, \qquad v \in \mathcal{V}^{\mathrm{P}}_{\omega,r}. \tag{23}$$

For a fixed route configuration $(w, z_{\omega,r})$, all constraints in the following LP are imposed for route r. The route-level LP value is

$$\begin{aligned} Q^R_{\omega,r}(w, z_{\omega,r}) = \min_{y_{\omega,r}} \quad & C^{\mathrm{chg}}_{\omega,r}(y_{\omega,r}) \\ \text{s.t.} \quad & (11)\text{–}(19), \\ & (23). \end{aligned} \tag{24}$$

with value $+\infty$ if this route LP is infeasible. Define the scenario-level Phase 1 value by

$$Q^R_\omega(w, z_\omega) = \sum_{r\in\mathcal{R}} Q^R_{\omega,r}(w, z_{\omega,r}).$$

Replacing exact scenario recourse by the route-level LP values gives the following Phase 1 lower-bounding formulation:

$$\begin{aligned} V^R = \min_{w,z} \quad & \left[C^{\mathrm{fac}}(w) + \sum_{\omega\in\Omega} p_\omega C^{\mathrm{bat}}_\omega(z_\omega)\right] + \sum_{\omega\in\Omega} p_\omega \sum_{r\in\mathcal{R}} Q^R_{\omega,r}(w, z_{\omega,r}) \\ \text{s.t.} \quad & (2)\text{–}(3), \\ & (5)\text{–}(7), \qquad\qquad \omega \in \Omega. \end{aligned} \tag{25}$$

Lemma 1 (Validity of the Phase 1 relaxation)**.** *For any lifted design configuration (w, z_ω) satisfying the upper-level design constraints,*

$$Q^R_\omega(w, z_\omega) \le Q_\omega(w, z_\omega). \tag{26}$$

If any route-level Phase 1 LP is infeasible, the exact operational subproblem is infeasible.

Proof. If the exact operational problem is infeasible, then $Q_\omega(w, z_\omega) = +\infty$ and (26) holds in the extended-real sense. Otherwise, every feasible solution of the exact operational problem satisfies (23). After removing the constraint family (9), any q^{P}_v satisfying (23) can be extended to the remaining occupancy constraints by setting $x_{v\tau} = 1$ for all $\tau \in \mathcal{T}_v$. Thus every exact feasible solution maps to a feasible collection of route-level Phase 1 solutions with the same charging cost. Hence $Q^R_\omega(w, z_\omega) \le Q_\omega(w, z_\omega)$, and, by contraposition, infeasibility of any route-level Phase 1 LP implies infeasibility of the exact operational problem. □

Because (25) and (22) have the same lifted design domain, Lemma 1 implies $V^R \le V^L$.

The Phase 1 formulation provides a valid lower bound but does not solve the exact lifted problem because it removes shared plug-in capacity coupling; after that removal, binary plug-in occupancy is projected out exactly. Phase 2 therefore restores the exact conditional recourse $Q_\omega(w, z_\omega)$ defined in (20). For each scenario configuration (w, z_ω) generated by the master, Phase 2 solves the exact scenario operational MIP to proven infeasibility or optimality. Thus, Phase 2 either certifies scenario operational infeasibility for (w, z_ω) or recovers its exact recourse value. The restricted masters, phase transition, and separation procedures are presented in Section 4.4.

4.3. Scenario-level resource monotonicity

We next establish a resource-monotonicity property for the exact scenario operational value Q_ω. As in (20), set $Q_\omega(w, z_\omega) = +\infty$ when the exact operational MIP is infeasible. Under the operational model in Section 3, trip-level energy requirements are known and do not vary with the selected battery capacity.

Let $\mathcal{W}$ collect all candidate wireless-facility indices, covering both SWC stations and DWC links, and let w_j denote the corresponding binary facility variable. Plug-in charger capacities are fixed shared resources in Section 3, not planning variables, and therefore are not included in $\mathcal{W}$. For two scenario configurations satisfying the lifted-master design constraints (2)–(3) and the battery-selection constraints (5)–(7), write $(w, z_\omega) \preceq_\omega (\bar{w}, \bar{z}_\omega)$ if

$$w_j \le \bar{w}_j, \quad j \in \mathcal{W}, \qquad B_{\omega,r}(z_{\omega,r}) \le B_{\omega,r}(\bar{z}_{\omega,r}), \quad r \in \mathcal{R}.$$

Proposition 1 (Scenario-level resource monotonicity)**.** *For a fixed scenario $\omega \in \Omega$, if $(w, z_\omega) \preceq_\omega (\bar{w}, \bar{z}_\omega)$, then*

$$Q_\omega(\bar{w}, \bar{z}_\omega) \le Q_\omega(w, z_\omega).$$

Proof. If $Q_\omega(w, z_\omega) = +\infty$, the inequality is immediate. Otherwise, take any feasible exact operating solution under (w, z_ω). Keep all plug-in occupancy and charging decisions unchanged. The plug-in constraints remain feasible because their variables and fixed capacities are unchanged, while the SWC and DWC charging bounds remain valid because the corresponding facility variables weakly increase.

For each route r, let

$$B_{\omega,r} = B_{\omega,r}(z_{\omega,r}), \qquad \bar{B}_{\omega,r} = B_{\omega,r}(\bar{z}_{\omega,r}), \qquad \Delta_r = \bar{B}_{\omega,r} - B_{\omega,r} \ge 0.$$

Increase every arrival and departure energy state on that route by $\overline{\rho}\Delta_r$. Because trip-level energy requirements are independent of battery capacity, this common shift preserves all station and link energy-balance equations. Moreover, if $\underline{\rho} B_{\omega,r} \le e \le \overline{\rho} B_{\omega,r}$, then, since $\overline{\rho} \ge \underline{\rho}$,

$$\underline{\rho}\bar{B}_{\omega,r} \le e + \overline{\rho}\Delta_r \le \overline{\rho}\bar{B}_{\omega,r}.$$

The same shift preserves the initial boundary equality and the return lower bound. It also preserves the overnight balance because both $\overline{\rho}B_{\omega,r}$ and the terminal departure energy increase by $\overline{\rho}\Delta_r$.

Hence the transformed solution is feasible under $(\bar{w}, \bar{z}_\omega)$ and has the same operational objective value. Minimization therefore gives $Q_\omega(\bar{w}, \bar{z}_\omega) \le Q_\omega(w, z_\omega)$. □

This monotonicity concerns Q_ω only and does not imply monotonicity of $C^{\text{bat}}_\omega(z_\omega) + Q_\omega(w, z_\omega)$, because larger batteries may have higher battery cost.

Corollary 1 (Scenario-level dominance implications)**.** *For any scenario configuration (w^*, z^*_ω) satisfying the same upper-level constraints, Proposition 1 implies:*

$$Q_\omega(w^*, z^*_\omega) = +\infty,\ (w, z_\omega) \preceq_\omega (w^*, z^*_\omega) \Rightarrow Q_\omega(w, z_\omega) = +\infty, \tag{27}$$

$$Q_\omega(w^*, z^*_\omega) < +\infty,\ (w, z_\omega) \preceq_\omega (w^*, z^*_\omega) \Rightarrow Q_\omega(w, z_\omega) \ge Q_\omega(w^*, z^*_\omega). \tag{28}$$

Phase 2 in Section 4.4 uses these implications to strengthen pointwise scenario cuts.

4.4. Complete solution procedure

The complete method is a two-phase integer L-shaped procedure. Phase 1 generates transferable route-level lower bounds through LP dual and Farkas separation. Phase 2 retains these bounds and performs exact scenario-level separation strengthened by scenario-level resource monotonicity.

4.4.1. Phase 1: Route-level lower-bounding separation

Phase 1 replaces each $Q^R_{\omega,r}(w, z_{\omega,r})$ in (25) with a proxy $\theta_{\omega,r}$. Let $\mathcal{F}^{R,\text{cur}}_{\omega,r}$ and $\mathcal{O}^{R,\text{cur}}_{\omega,r}$ denote the current route-level feasibility and optimality cut pools, and write $\theta^R = \{\theta_{\omega,r}\}_{\omega\in\Omega, r\in\mathcal{R}}$. The Phase 1 restricted master is

$$\begin{aligned}
\min_{w,z,\theta^R} \quad & C^{\text{fac}}(w) + \sum_{\omega\in\Omega} p_\omega \left[C^{\text{bat}}_\omega(z_\omega) + \sum_{r\in\mathcal{R}} \theta_{\omega,r} \right] & & \\
\text{s.t.} \quad & (2)\text{–}(3), & & \\
& (5)\text{–}(7), & & \omega \in \Omega, \\
& \theta_{\omega,r} \ge 0, & & \omega \in \Omega,\ r \in \mathcal{R}, \\
& \psi^R_{\omega,r}(w, z_{\omega,r}) \ge 0, & & \omega \in \Omega,\ r \in \mathcal{R},\ \psi^R_{\omega,r} \in \mathcal{F}^{R,\text{cur}}_{\omega,r}, \\
& \theta_{\omega,r} \ge \varphi^R_{\omega,r}(w, z_{\omega,r}), & & \omega \in \Omega,\ r \in \mathcal{R},\ \varphi^R_{\omega,r} \in \mathcal{O}^{R,\text{cur}}_{\omega,r}.
\end{aligned} \tag{29}$$

At each master solution, a feasible route LP generates a standard LP dual optimality cut, while an infeasible route LP generates a standard Farkas feasibility cut; Appendix Appendix B.1 gives the explicit cut forms. Validity of the route cuts follows from LP duality and the Farkas certificate in Appendix Appendix B.1. The theoretical Phase 1 loop repeatedly solves (29) and separates route-level cuts until no violated Phase 1 cut remains.

4.4.2. Phase 2: Exact scenario separation

At the phase transition, the route proxies and route cut pools are retained. Phase 2 introduces one scenario-level proxy θ_ω per scenario and links the retained route lower bounds in the restricted master. Let $\mathcal{F}_\omega$ and $\mathcal{O}_\omega$ denote the complete scenario-level feasibility and optimality cut families, with current finite subsets $\mathcal{F}_\omega^{S,\mathrm{cur}}$ and $\mathcal{O}_\omega^{S,\mathrm{cur}}$. Because operational costs are nonnegative, use $L_\omega = 0$ as a valid global lower bound on operational recourse. Writing $\theta^S = \{\theta_\omega\}_{\omega\in\Omega}$, the Phase 2 restricted master is

$$
\begin{aligned}
\min_{w,z,\theta^R,\theta^S} \quad & C^{\mathrm{fac}}(w) + \sum_{\omega\in\Omega} p_\omega \left[C_\omega^{\mathrm{bat}}(z_\omega) + \theta_\omega \right] \\
\text{s.t.} \quad & (2)\text{–}(3), \\
& (5)\text{–}(7), && \omega \in \Omega, \\
& \theta_{\omega,r} \geq 0, && \omega \in \Omega,\ r \in \mathcal{R}, \\
& \psi_{\omega,r}^R(w, z_{\omega,r}) \geq 0, && \omega \in \Omega,\ r \in \mathcal{R},\ \psi_{\omega,r}^R \in \mathcal{F}_{\omega,r}^{R,\mathrm{cur}}, \\
& \theta_{\omega,r} \geq \varphi_{\omega,r}^R(w, z_{\omega,r}), && \omega \in \Omega,\ r \in \mathcal{R},\ \varphi_{\omega,r}^R \in \mathcal{O}_{\omega,r}^{R,\mathrm{cur}}, \\
& \theta_\omega \geq \sum_{r\in\mathcal{R}} \theta_{\omega,r}, && \omega \in \Omega, \\
& \theta_\omega \geq L_\omega, && \omega \in \Omega, \\
& \psi_\omega^S(w, z_\omega) \geq 0, && \omega \in \Omega,\ \psi_\omega^S \in \mathcal{F}_\omega^{S,\mathrm{cur}}, \\
& \theta_\omega \geq \varphi_\omega^S(w, z_\omega), && \omega \in \Omega,\ \varphi_\omega^S \in \mathcal{O}_\omega^{S,\mathrm{cur}}.
\end{aligned}
\tag{30}
$$

For a Phase 2 master candidate $(w, z, \theta^R, \theta^S)$, exact separation solves (20) for each scenario. The baseline exact separation uses pointwise integer L-shaped cuts. For an evaluated scenario configuration (w^*, z_ω^*), let $k_{\omega,r}^*$ be the selected battery option on route r and define the pointwise mismatch expression

$$
D_\omega(w, z_\omega; w^*, z_\omega^*) = \sum_{j\in\mathcal{W}: w_j^*=0} w_j + \sum_{j\in\mathcal{W}: w_j^*=1} (1 - w_j) + \sum_{r\in\mathcal{R}} \left(1 - z_{\omega,r,k_{\omega,r}^*}\right).
$$

The pointwise exact cuts are

$$
\begin{aligned}
D_\omega(w, z_\omega; w^*, z_\omega^*) \geq 1, & && \text{if } Q_\omega^* = +\infty, \\
\theta_\omega \geq Q_\omega^* - (Q_\omega^* - L_\omega) D_\omega(w, z_\omega; w^*, z_\omega^*), & && \text{if } Q_\omega^* < +\infty.
\end{aligned}
$$

The first cut is a no-good feasibility cut that excludes only the evaluated infeasible scenario configuration. The second is a pointwise integer L-shaped optimality cut that restores the exact recourse value only at the evaluated feasible scenario configuration.

The full method uses the scenario-level dominance implications in Corollary 1 to extend a pointwise correction to the dominated resource region. Define the scenario-level monotonicity mismatch expression

$$
I_\omega(w, z_\omega; w^*, z_\omega^*) = \sum_{j\in\mathcal{W}: w_j^*=0} w_j + \sum_{r\in\mathcal{R}} \sum_{k\in\mathcal{K}: \beta_k > \beta_{k_{\omega,r}^*}} z_{\omega,r,k}.
$$

The monotonicity-strengthened scenario cuts are

$$
\begin{aligned}
I_\omega(w, z_\omega; w^*, z_\omega^*) \geq 1, & && \text{if } Q_\omega^* = +\infty, \\
\theta_\omega \geq Q_\omega^* \left[1 - I_\omega(w, z_\omega; w^*, z_\omega^*)\right], & && \text{if } Q_\omega^* < +\infty.
\end{aligned}
$$

The strengthened feasibility cut excludes the evaluated infeasible configuration and every dominated configuration with no more wireless or battery resources. The strengthened optimality cut imposes the recourse lower bound Q_ω^* throughout the same dominated region. Validity follows from the scenario-level dominance implications in Corollary 1 and the global lower bound L_ω. Since $I_\omega \leq D_\omega$, the strengthened feasibility cut weakly dominates the pointwise no-good feasibility cut, and the strengthened optimality cut weakly dominates the pointwise optimality cut.

Before exact separation, an enabled RSP prescreen may add globally valid LP cuts; omitting RSP does not affect exactness of Phase 2.

4.4.3. Algorithm, bounds, and convergence

When every scenario exact operational MIP is proven feasible and solved to optimality for a restricted-master candidate (w, z), its candidate upper bound is

$$UB^{\text{cand}}(w, z) = C^{\text{fac}}(w) + \sum_{\omega \in \Omega} p_\omega C^{\text{bat}}_\omega(z_\omega) + \sum_{\omega \in \Omega} p_\omega Q_\omega(w, z_\omega). \tag{31}$$

Algorithm 1 summarizes the full method.

Input : Scenario data, candidate wireless facilities, candidate battery options
Output: Facility layout w, scenario battery configurations z, lower bound LB, upper bound UB

Initialize route-level and scenario-level cut pools; set $LB \leftarrow -\infty$ and $UB \leftarrow +\infty$;

`// Phase 1: route-level lower-bounding separation`
repeat
  Solve the Phase 1 restricted master to optimality and set $LB \leftarrow \max\{LB, \text{master bound}\}$;
  Solve route-level LP relaxations for each scenario at the current (w, z_ω);
  Add violated route-level dual and Farkas cuts;
until *no violated route-level Phase 1 cut is found*;
Transfer route proxies and cut pools; initialize the Phase 2 restricted master;

`// Phase 2: exact mixed-integer verification`
repeat
  Solve the restricted master (30) to optimality and set $LB \leftarrow \max\{LB, \text{master bound}\}$;
  If enabled, add violated RSP cuts;
  if *an RSP cut is added* **then**
    continue to the next master solve;
  end
  Solve exact scenario operational MIPs at the current (w, z_ω) to proof;
  if *one or more scenarios are proven infeasible* **then**
    Add exact scenario feasibility corrections;
    continue to the next master solve;
  end
  if *all scenario MIPs are proven optimal* **then**
    Compute UB^{cand} using (31);
    Set $UB \leftarrow \min\{UB, UB^{\text{cand}}\}$ and update the incumbent if improved;
    foreach *scenario whose exact recourse is underestimated by* θ_ω **do**
      Add exact scenario optimality corrections;
    end
  end
until *no violated Phase 2 cut is found*;
return *Best incumbent, LB, and UB*;

Algorithm 1: Full reformulation-enabled integer L-shaped method

When a finite incumbent is available, the implementation reports the relative gap $(UB - LB)/\max\{1, |UB|\}$; otherwise it reports $+\infty$.

Theorem 1 (Finite convergence under exact separation)**.** *Assume that $p_\omega > 0$ for all $\omega \in \Omega$, the binary facility–battery configuration space for $(w, \{z_\omega\}_{\omega \in \Omega})$ is finite, the Phase 2 restricted master is initially feasible and bounded, and the lifted formulation has a finite optimum. Assume that Phase 1 terminates finitely, optional RSP adds only finitely many globally valid cuts, all generated cuts are globally valid and retained, and exact separation produces a feasibility or optimality correction whenever the current scenario configuration is infeasible or has underestimated recourse. The theoretical algorithm solves every restricted master and exact Phase 2 scenario subproblem to proof. Then the exact Phase 2 separation loop terminates finitely with an optimal solution for the lifted reformulation and hence for the original two-stage formulation.*

Proof. Global validity makes each restricted master a relaxation of the exact lifted formulation, so its optimal value is a valid lower bound. Whenever all exact scenario subproblems are proven feasible and solved to optimality, (31) gives a valid upper bound. If a restricted-master optimizer is not yet exact, exact separation adds a retained correction that either excludes an infeasible scenario configuration or enforces its exact recourse value. Such a corrected finite binary configuration cannot require the same exact correction again; hence only finitely many exact corrections can occur.

At termination, each current scenario configuration is exact-feasible. No current θ_ω underestimates $Q_\omega(w, z_\omega)$. Because the master minimizes θ_ω with $p_\omega > 0$, and all retained constraints imposing lower bounds on θ_ω do not exceed $Q_\omega(w, z_\omega)$, $\theta_\omega = Q_\omega(w, z_\omega)$ for all scenarios in the current complete configuration. The restricted-master objective then equals the exact candidate value. Thus $LB = UB$ at termination. The lifted-reformulation equivalence in Section 4.1 gives optimality for the original two-stage formulation. □

The theorem describes the theoretical algorithm with finite Phase 1, finite valid RSP cuts, optimal restricted-master solves, and proven exact scenario MIPs. In the time-limited implementation, Phase 1 may switch under an implementation stopping rule, the solver best bound updates LB, and the master incumbent is used for separation. If an exact scenario MIP is not solved to proof before the global time limit, it is not used to generate an exact cut, update the exact upper bound, or certify optimal termination. The implementation then returns the current incumbent, lower bound, and reported gap.

5. Computational Results

This section evaluates the proposed reformulation and solution method on Tower-derived wireless charging planning instances. The experiments use the bus network as a realistic source of shared stops, shared links, and route-level service plans, while the main purpose of the section is to validate algorithmic performance.

5.1. Experimental design and instance families

The test instances are derived from the Tower Transit bus network in Singapore's Jurong area. The resulting data set provides a realistic spatial structure with shared stops, shared road links, and multiple routes that may use the same wireless charging infrastructure. We use these instances as a computational test bed rather than as a public benchmark.

Each scenario represents an alternative medium-term service plan for the same physical network, not random same-day scheduling noise. Scenario-dependent data include the operated routes, bus duties, timetables, energy requirements, and charging opportunities induced by the corresponding service plan. The first-stage wireless charging facilities are shared across all scenarios, whereas battery and charging-operation decisions adapt after the service plan is realized.

We use a compact instance code of the form `N-k`, where `N` indicates the network scale and k indicates the number of service-plan scenarios. Specifically, `S`, `M`, `B`, and `L` denote the small, medium, base, and large Tower-derived networks, respectively. The capacity-coupling experiment appends `Cq` to the base 10-scenario instance, where q indexes the number of existing plug-in fast chargers. The network-scale family fixes the scenario count at three and varies the physical network scale. The scenario-count family fixes the base network and varies the number of service-plan scenarios. Motivated by the relaxation-verification structure in Section 4.2, the capacity-coupling family varies plug-in fast-charger capacity on `B-10` to change how strongly residual fast-charging demand couples route-level recourse decisions. Table 1 summarizes the resulting instance families and their experimental roles.

Travel times are computed from link distances using a nominal operating speed of 30 km/h and then rounded to the one-minute scheduling grid.[1] The remaining parameters specify battery costs, charging rates, charging efficiencies, electricity prices, facility costs, and state-of-charge limits. Table 2 reports the main parameter values used in the computational experiments.

The benchmark ladder consists of five methods: the direct extensive-form MILP, baseline integer L-shaped decomposition, lifted integer L-shaped decomposition, lifted two-phase method, and the full method. The direct extensive-form MILP optimizes the deterministic equivalent with a commercial solver. The baseline integer L-shaped decomposition applies the classical scenario decomposition induced by the original two-stage model. The lifted integer L-shaped decomposition uses the lifted master representation without the two-phase relaxation. Lifted two-phase adds the Phase 1 lower-bounding and Phase 2 verification structure. The full method further adds scenario-level monotonicity-based strengthened cuts.

Table 1: Main instance families and experimental use

Instance family	Data source or variant	Varied dimension	Primary use	Compared methods
`S-3`, `M-3`, `B-3`, `L-3`	`tower_small`, `tower_medium`, `tower`, `large`	Network scale, with three scenarios fixed	Network-scale ablation	Direct MILP, baseline integer L-shaped, lifted integer L-shaped, lifted two-phase, full method
`B-k`, $k \in \{3, 5, 10, 20, 40\}$	`tower`, `tower_5`, `tower_10`, `tower_20`, `tower_40`	Scenario count, with base network fixed	Scenario-count scalability	Direct MILP, lifted two-phase, full method
`B-10-Cq`, $q \in \{1, 2, 3, 4, 5\}$	Capacity variants of `tower_10`	Existing plug-in fast-charger units, with base network and 10 scenarios fixed	Relaxation-gap stress test	Lifted two-phase, full method

`B-3` is the representative base instance used for solution interpretation; larger q indicates more existing plug-in fast-charger units.

5.2. Solution characteristics on the base Tower instance

This subsection will use `B-3` as the representative planning solution.

The solution-summary table will report selected SWC facilities, selected DWC links, cost components, objective value, and battery-capacity summaries.

Representative selected stops or links will show that the model places wireless infrastructure on shared parts of the bus network.

The planning interpretation is that the model recommends a network-level wireless charging layout under service-plan uncertainty rather than a single-day charging schedule.

[1] The one-minute lower bound avoids zero-duration movements on short links and absorbs acceleration, deceleration, signal delay, and station-approach time.

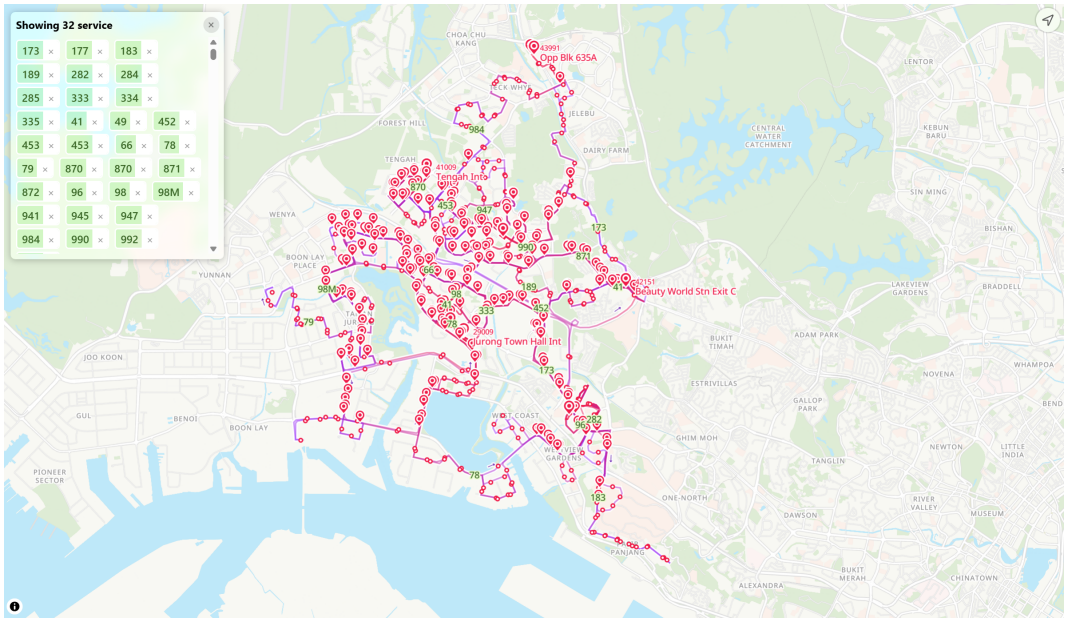


Figure 1: Spatial layout of the base Tower-derived network in the Jurong area.

Table 2: Parameter settings for the computational experiments

Parameter	Description	Value	Source
Battery and state of charge			
c^{bat}	Daily-equivalent battery capacity cost	0.148 USD/kWh/day	BloombergNEF price amortized over Wang et al. battery life (BloombergNEF, 2025; Wang et al., 2024)
$\mathcal{K}_r$	Candidate route-level battery options	10-kWh increments within instance bounds	Case-data preprocessing
$\underline{\rho}, \overline{\rho}$	Minimum and maximum state-of-charge fractions	0.2, 0.8	Wireless bus planning literature (Li et al., 2024)
Charging operation			
r^{F}	Plug-in fast-charging rate	2.5 kWh/min	Najafi et al. (2025)
r^{SWC}	Stationary wireless charging rate	1.0 kWh/min	Wang et al. (2024)
r^{DWC}	Dynamic wireless charging rate	1.0 kWh/min	Wang et al. (2024)
η^{F}	Plug-in fast-charging efficiency	0.90	Li et al. (2024)
η^{SWC}	Stationary wireless charging efficiency	0.85	Sun et al. (2024)
η^{DWC}	Dynamic wireless charging efficiency	0.80	Sun et al. (2024)
a_s	Existing plug-in charger units at each fast-charging location	4 chargers	Case-study assumption
Infrastructure and electricity cost			
T^{fac}	Facility amortization horizon	30 years	Wang et al. (2024)
c_j^{fac}, $j \in \mathcal{J}^{\text{SWC}}$	Daily-equivalent cost of an SWC facility	2.16 USD/day	Wang et al. (2024)
c_j^{fac}, $j \in \mathcal{J}^{\text{DWC}}$	Daily-equivalent cost of a DWC link of length d_j km	$9d_j + 0.90$ USD/day	Wang et al. (2024)
c^{day}	Daytime electricity price	0.1922 USD/kWh	Singapore tariff data (Singapore Department of Statistics, 2026)
c^{night}	Nighttime electricity price	0.1233 USD/kWh	Singapore tariff data (Singapore Department of Statistics, 2026)

5.3. Network-scale algorithmic ablation

The network-scale ablation compares five methods on S-3, M-3, B-3, and L-3.

The largest reported instances are intended to show where direct optimization and simpler decomposition variants stop being computationally effective.

Table 3: Network-scale ablation under three service-plan scenarios

Instance	Direct MILP	Baseline integer L-shaped	Lifted integer L-shaped	Lifted two-phase	Full method
S-3	Conv., 4.0s	TL, 98.3%	TL, 80.0%	Conv., 0.3s	Conv., 0.4s
M-3	TL, 0.28%	skipped	skipped	Conv., 26.2s	Conv., 21.6s
B-3	OOM, 0.27%	skipped	skipped	TL, 0.60%	Conv., 45.4s
L-3	TODO	skipped	skipped	TODO	TODO

Conv. = converged within the 0.01% tolerance; TL = 7200s time limit with incumbent;
OOM = Gurobi memory error 10001. Reported percentages are final relative gaps. Prior-run values are placeholders to be rerun.

The mechanism contrast is that lifted two-phase still generates many weak verification cycles on B-3, whereas the strengthened full method converges quickly with only a few Phase 2 iterations.

Table 4: Mechanism contrast on B-3

Method	Status	Gap (%)	Time (s)	Iterations
Lifted two-phase	TL	0.595	7211.8	548
Full method	Conv.	0.010	45.4	9/3

Full-method iterations are reported as Phase 1/Phase 2 iterations.

The phase-level evidence supports the interpretation that the full method controls both first-stage bound improvement and second-stage mixed-integer verification.

5.4. Scenario-count scalability

This subsection uses the B-k family, with $k \in \{3, 5, 10, 20, 40\}$, to test scenario-count scalability on the base Tower-derived network. The comparison is limited to the direct extensive-form MILP, lifted two-phase method, and full method because the simpler integer L-shaped variants already fail in the network-scale ablation.

Table 5: Scenario-count scalability on the base Tower-derived network

Instance	Direct MILP	Lifted two-phase	Full method	Phase 1/2 (s)	Iters P1/P2
B-3	OOM, 0.27%	TL, 0.60%	Conv., 45.4s	29.7/15.7	9/3
B-5	TODO	TODO	Conv., 52.8s	36.3/16.5	6/3
B-10	TODO	TODO	Conv., 126.6s	68.1/58.5	3/4
B-20	TODO	TODO	Conv., 356.3s	145.6/210.7	9/4
B-40	TODO	TODO	Conv., 1702.1s	447.3/1254.8	12/2

All entries are placeholders from prior runs and should be rerun under the final settings.

The main diagnostic is how total time, Phase 1 time, Phase 2 time, and iteration counts change as more scenario-specific recourse problems are added to the same first-stage infrastructure plan.

5.5. Capacity-coupling and relaxation-gap experiment

This subsection uses B-10-C1 through B-10-C5 to test the effect of shared plug-in fast-charger capacity on the relaxation-verification structure. Within this subsection, we denote these five variants by C1–C5; all five are built from B-10, and only the number of existing plug-in fast chargers is varied. This changes how residual fast-charging demand, after accounting for wireless charging opportunities and battery choices, couples the route-level recourse problems. The experiment is designed around the Phase 1 relaxation in Lemma 1, which removes shared plug-in fast-charger capacity coupling to obtain route-level LP subproblems. When capacity is tight, the Phase 1 relaxation is expected to be less aligned with Phase 2; when capacity is loose, the two feasible regions should be closer. The comparison is restricted to lifted two-phase and the full method because both use the same lifted reformulation, Phase 1 relaxation, and exact Phase 2 verification; the main difference is the monotonicity strengthening. The main computational question is how the relaxation gap and Phase 2 verification burden change with capacity, and whether monotonicity strengthening remains effective as capacity coupling becomes tighter.

Table 6: Capacity-coupling and relaxation-gap stress test on B-10

Instance	Lifted two-phase	Full method	Initial P2 gap	P1 time	P2 master time	Exact subproblem time	P1/P2 cuts
C1	TODO	TODO	TODO	TODO	TODO	TODO	TODO
C2	TODO	TODO	TODO	TODO	TODO	TODO	TODO
C3	TODO	TODO	TODO	TODO	TODO	TODO	TODO
C4	TODO	TODO	TODO	TODO	TODO	TODO	TODO
C5	TODO	TODO	TODO	TODO	TODO	TODO	TODO

All entries are placeholders for the final capacity-coupling experiment.

6. Conclusion

This paper studies a two-stage wireless charging facility-location problem for electric-bus systems under service-plan uncertainty. The model treats SWC and DWC deployment as strategic here-and-now decisions, while route-level battery selection and representative-day charging operations adapt after each service-plan scenario is realized. The resulting recourse structure captures how wireless charging opportunities affect battery requirements and residual demand for limited shared plug-in fast chargers.

The main methodological contribution is an exact decomposition framework for this shared-resource mixed-integer recourse structure. The method uses an exact lifted reformulation that keeps scenario-dependent battery choices adaptive, a nested scenario–route relaxation–verification decomposition, and scenario-level resource monotonicity cuts. Phase 1 route-level LP relaxations obtain valid lower bounds and cuts, Phase 2 restores exact mixed-integer operational verification, and the strengthened full method applies monotonicity cuts to exact scenario recourse. Under exact master and recourse separation, the complete procedure converges to an optimal solution of the original two-stage formulation.

The computational study is designed to show how each algorithmic component contributes to scalability. The base planning instance illustrates the type of wireless facility layout and adaptive battery decisions produced by the model. The ablation, scenario-scaling, and capacity-coupling experiments then evaluate when the direct extensive-form MILP and simpler decomposition variants become insufficient and when the full method remains effective.

Several limitations define natural extensions. The current experiments use constructed service-plan scenarios, treat plug-in fast-charging infrastructure as exogenous, and do not endogenize the feedback from battery weight to route-level energy consumption. Future work can develop richer scenario-generation models, jointly optimize wireless and plug-in charging expansion, incorporate operational uncertainty, and apply the lifted two-phase decomposition idea to other stochastic infrastructure-planning problems with shared-resource recourse.

Appendix A. Model Notation

Appendix B. Proofs

Appendix B.1. Route-Level LP Cuts for Phase 1

For a fixed route pair (ω, r), write the Phase 1 route LP in standard form as

$$Q^R_{\omega,r}(w, z_{\omega,r}) = \min_{y_{\omega,r}\geq 0} \left\{ c^\top_{\omega,r} y_{\omega,r} : A_{\omega,r} y_{\omega,r} \geq b_{\omega,r} + H_{\omega,r} w + G_{\omega,r} z_{\omega,r} \right\}.$$

If the route LP is feasible, any dual-feasible vector

$$\pi_{\omega,r} \geq 0, \qquad A^\top_{\omega,r} \pi_{\omega,r} \leq c_{\omega,r},$$

gives the route-level optimality cut

$$\theta_{\omega,r} \geq \pi^\top_{\omega,r} \left(b_{\omega,r} + H_{\omega,r} w + G_{\omega,r} z_{\omega,r} \right).$$

If the route LP is infeasible at an evaluated configuration $(w^*, z^*_{\omega,r})$, a Farkas ray can be selected from

$$\Sigma^{\mathrm{F}}_{\omega,r} = \left\{ \sigma_{\omega,r} \geq 0 : A^\top_{\omega,r} \sigma_{\omega,r} \leq 0 \right\}$$

such that

$$\hat{\sigma}^\top_{\omega,r} \left(b_{\omega,r} + H_{\omega,r} w^* + G_{\omega,r} z^*_{\omega,r} \right) > 0.$$

Then any configuration with a feasible Phase 1 route relaxation must satisfy

$$\hat{\sigma}^\top_{\omega,r} \left(b_{\omega,r} + H_{\omega,r} w + G_{\omega,r} z_{\omega,r} \right) \leq 0. \tag{B.1}$$

Thus (B.1) is a valid feasibility cut for the Phase 1 relaxation.

Table A.7: Summary of notation

Symbol	Description
Sets and indices	
Ω	Set of service-plan scenarios, indexed by ω
$\mathcal{S}$	Set of stations, indexed by s
$\mathcal{L}$	Set of directed links between consecutive stations, indexed by l
C	Set of existing interchanges or hubs equipped with plug-in chargers, $C \subseteq \mathcal{S}$
$\mathcal{R}$	Set of routes, indexed by r
$\mathcal{K}$	Set of candidate battery-capacity options, indexed by k
$\mathcal{M}_{\omega,r}$	Set of bus trip chains on route r under scenario ω, indexed by m
$\mathcal{V}_{\omega,r}$	Set of station visits on route r under scenario ω, indexed by v
$\mathcal{V}^{\mathrm{P}}_{\omega,r}$	Set of plug-in charging visits on route r under scenario ω
$\mathcal{V}^{\mathrm{S}}_{\omega,r}$	Set of SWC-eligible station visits on route r under scenario ω
$\mathcal{A}_{\omega,r}$	Set of directed link traversals on route r under scenario ω, indexed by a
$\mathcal{T}_{\omega}$	Relevant plug-in charging time periods under scenario ω
$\mathcal{T}_{v}$	Time periods during which plug-in charging visit v can occupy a plug-in charger
Parameters	
p_{ω}	Probability of scenario ω
c^{S}_{s}	Representative-day equivalent cost of installing SWC at station s
c^{D}_{l}	Representative-day equivalent cost of installing DWC on link l
c^{bat}	Representative-day equivalent battery capacity cost per kWh
$c^{\mathrm{day}}, c^{\mathrm{night}}$	Daytime and overnight electricity prices
$n_{\omega,r}$	Fleet-size requirement for route r under scenario ω
β_k	Battery capacity value represented by option k
d_a	Energy consumption associated with directed link traversal a
d^{0}_{m}	Boundary energy consumption from the depot to the first station visit of trip chain m
d^{N}_{m}	Boundary energy consumption from the terminal station visit of trip chain m back to the depot
U_s	Number of existing plug-in chargers at station s
κ	Grid-side energy delivered by one occupied plug-in time slot
$\bar{q}^{\mathrm{S}}_{v}$	Maximum SWC energy available during station visit v
$\bar{q}^{\mathrm{D}}_{a}$	Maximum DWC energy available during directed link traversal a
$\eta^{\mathrm{P}}, \eta^{\mathrm{S}}, \eta^{\mathrm{D}}$	Charging efficiencies for plug-in charging, SWC, and DWC
$\underline{\rho}, \overline{\rho}$	Minimum and maximum state-of-charge fractions
Derived quantities and functions	
$s(v)$	Physical station associated with station visit v
$\ell(a)$	Directed link associated with traversal a
$\mathrm{tail}(a)$	Departure station visit of traversal a
$\mathrm{head}(a)$	Arrival station visit of traversal a
v^{0}_{m}	First station visit of trip chain m
v^{N}_{m}	Terminal station visit of trip chain m
Decision variables	
w^{S}_{s}	1 if SWC is installed at station $s \in \mathcal{S} \setminus C$, 0 otherwise
w^{D}_{l}	1 if DWC is installed on link $l \in \mathcal{L}$, 0 otherwise
$z_{\omega,r,k}$	1 if route r uses battery capacity option k in scenario ω, 0 otherwise
$B_{\omega,r}$	Selected battery capacity for route r in scenario ω
x_{ω}	Vector of scenario-level plug-in charger occupancy decisions in scenario ω
y_{ω}	Vector of scenario-level charging amounts and battery energy-state decisions in scenario ω
q^{P}_{v}	Plug-in charging energy assigned during plug-in visit v
q^{S}_{v}	SWC energy assigned during SWC-eligible station visit v
q^{D}_{a}	DWC energy assigned to directed link traversal a
q^{N}_{m}	Overnight charging energy assigned after trip chain m returns to the depot
$e^{\mathrm{arr}}_{v}, e^{\mathrm{dep}}_{v}$	Battery energy state at arrival and departure of station visit v
$x_{v\tau}$	1 if plug-in charging visit v occupies one plug-in charger at time τ, 0 otherwise

Appendix C. Additional Computational Results

References

BloombergNEF, 2025. Lithium-ion battery pack prices fall to $108 per kilowatt-hour, despite rising metal prices: Bloombergnef. BloombergNEF press release.

Li, W., He, Y., Hu, S., He, Z., Ratti, C., 2024. Planning dynamic wireless charging infrastructure for battery electric bus systems with the joint optimization of charging scheduling. Transportation Research Part C: Emerging Technologies 159, 104469. doi:10.1016/j.trc.2023.104469.

Najafi, A., Gao, K., Parishwad, O., Tsaousoglou, G., Jin, S., Yi, W., 2025. Integrated optimization of charging infrastructure, electric bus scheduling and energy systems. Transportation Research Part D: Transport and Environment 141, 104664. doi:10.1016/j.trd.2025.104664.

Singapore Department of Statistics, 2026. Electricity tariff, annual. URL: https://data.gov.sg/datasets/d_d610f8ed1864daa6c7e790318bbc3323/view.

Sun, R., Luo, Q., Chen, Y., 2024. Optimizing dynamic wireless charging for electric buses: A data-driven approach to infrastructure planning. Applied Energy 373, 123912. doi:10.1016/j.apenergy.2024.123912.

Wang, Y., Zhou, Y., Yan, X., 2024. Reliable dynamic wireless charging infrastructure deployment problem for public transport services. European Journal of Operational Research 313, 747–766. doi:10.1016/j.ejor.2023.08.053.